

A - Appender

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 100 points

Problem Statement

You are given a string S consisting of lowercase English letters.

Output the string T determined as follows.

- If the last character of S is e, T is the string obtained by appending r to the end of S .
- If the last character of S is not e, T is the string obtained by appending er to the end of S .

Constraints

- S is a string consisting of lowercase English letters.
- The length of S is between 1 and 10, inclusive.

Input

The input is given from Standard Input in the following format:

S

Output

Output the answer.

Sample Input 1

live

Sample Output 1

liver

The last character of live is e, so the string liver, obtained by appending r to the end of live, is the string T to be outputted.

Sample Input 2

```
femur
```

Sample Output 2

```
femurer
```

The last character of femur is not e, so the string femurer, obtained by appending er to the end of femur, is the string T to be outputted.

Sample Input 3

```
chimpanzee
```

Sample Output 3

```
chimpanzeer
```

The last character of chimpanzee is e, so the string chimpanzeer, obtained by appending r to the end of chimpanzee, is the string T to be outputted.

B - Wild Card

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 200 points

Problem Statement

S is a string of length N consisting of lowercase English letters.

T is a string of length N consisting of lowercase English letters and $*$.

We say that S matches T if and only if T can be made equal to S by replacing each $*$ in T with some lowercase English letter.

Determine whether S matches T .

Constraints

- $1 \leq N \leq 100$
- S is a string of length N consisting of lowercase English letters.
- T is a string of length N consisting of lowercase English letters and $*$.

Input

The input is given from Standard Input in the following format:

```
 $N$   
 $S$   
 $T$ 
```

Output

Output Yes if S matches T , and No otherwise, in one line.

Sample Input 1

```
8  
chokudai  
**o*u*ai
```

Sample Output 1

```
Yes
```

By replacing the first, second, fourth, and sixth characters of T , which are *, with c, h, k, and d, respectively, T can be made equal to S .

Sample Input 2

```
5
snuke
snake
```

Sample Output 2

```
No
```

Sample Input 3

```
5
yiwiy
*****
```

Sample Output 3

```
Yes
```

C - Third Largest Number

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 300 points

Problem Statement

You are given an integer N not less than 3 and a length- N sequence of positive integers $A = (A_1, A_2, \dots, A_N)$.

For each $k = 3, 4, \dots, N$, solve the following problem:

- When $A_1, A_2, \dots, A_k$ are sorted in descending order, find the third value from the front.

Constraints

- $3 \leq N \leq 5 \times 10^5$
- $1 \leq A_i \leq 10^9$
- All input values are integers.

Input

The input is given from Standard Input in the following format:

```
N
A_1 A_2 ... A_N
```

Output

Output $N - 2$ lines.

The i -th line ($1 \leq i \leq N - 2$) should contain the answer for $k = i + 2$.

Sample Input 1

```
5
1 2 1 2 3
```

Sample Output 1

```
1
1
2
```

- For $k = 3$: sorting A_1, A_2, A_3 in descending order gives 2, 1, 1. The third value from the front is 1, so output 1 on the first line.
- For $k = 4$: sorting A_1, A_2, A_3, A_4 in descending order gives 2, 2, 1, 1. The third value from the front is 1, so output 1 on the second line.
- For $k = 5$: sorting A_1, A_2, A_3, A_4, A_5 in descending order gives 3, 2, 2, 1, 1. The third value from the front is 2, so output 2 on the third line.

Sample Input 2

```
10
1 1 1 3 2 5 4 3 6 5
```

Sample Output 2

```
1
1
1
2
3
3
4
5
```

Sample Input 3

```
10
11 9 1 3 17 19 10 19 17 3
```

Sample Output 3

```
1
3
9
11
11
17
17
17
```


D - Automat

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 425 points

Problem Statement

In the Kingdom of AtCoder, two kinds of bills are in circulation: 1-dollar bills and K -dollar bills.

At Company J's fully automated cafeteria in the kingdom, desserts and drinks are sold as products.

The company's fully automated cafeteria has a dessert vending machine that sells N desserts, and a drink vending machine that sells M drinks. The desserts are numbered 1 through N , and the drinks are numbered 1 through M .

Dessert i costs A_i dollars, and drink j costs B_j dollars.

As payment, the dessert vending machine accepts both 1-dollar bills and K -dollar bills, but the drink vending machine accepts only K -dollar bills. Both vending machines give change using only 1-dollar bills. You cannot buy two or more of the same product.

Takahashi came to Company J's fully automated cafeteria with X 1-dollar bills and Y K -dollar bills.

Among the combinations of products that he can buy with the bills he has, find the maximum number of products he can purchase.

Constraints

- $1 \leq N \leq 2 \times 10^5$
 - $1 \leq M \leq 2 \times 10^5$
 - $2 \leq K \leq 10^9$
 - $0 \leq X \leq 10^{15}$
 - $0 \leq Y \leq 10^9$
 - $1 \leq A_i \leq 10^9$ ($1 \leq i \leq N$)
 - $1 \leq B_j \leq 10^9$ ($1 \leq j \leq M$)
 - All input values are integers.
-

Input

The input is given from Standard Input in the following format:

```
 $N$   $M$   $K$   
 $X$   $Y$   
 $A_1$   $\dots$   $A_N$   
 $B_1$   $\dots$   $B_M$ 
```

Output

Output the answer.

Sample Input 1

```
2 3 10  
50 6  
22 30  
20 12 24
```

Sample Output 1

```
4
```

By shopping as follows, Takahashi can buy four products.

- Pay two 10-dollar bills to buy drink 1. No change.
- Pay two 10-dollar bills to buy drink 2. The change is eight 1-dollar bills.
- Pay two 10-dollar bills and ten 1-dollar bills to buy dessert 2. No change.
- Pay twenty-two 1-dollar bills to buy dessert 1. No change.

He cannot buy more than four products, so the answer is 4.

Sample Input 2

```
1 7 67  
677677677766666 0  
777666777  
20 12 24 67 67 67 67
```

Sample Output 2

```
1
```

By shopping as follows, Takahashi can buy one product.

- Pay 600000777666777 1-dollar bills to buy dessert 1. The change is 6000000000000000 1-dollar bills.

He cannot buy more than one product, so the answer is 1.

Sample Input 3

```
20 20 30
605776135 133105105
97363214 218434035 697895427 109255624 299037330 227873982 195540071 411713803 828357845 24
4535208 138059186 639510883 39844882 707397687 371274487 696536603 351588202 319490007 4712
1612 87169661
32256972 567982330 554885983 299718223 443859449 687952877 264684780 666659381 576335424 94
1894234 406248934 321334900 423472560 863738035 213143887 384834384 468161291 673106162 164
648316 15903323
```

Sample Output 3

```
22
```

E - Min-Max Swap

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 450 points

Problem Statement

You are given a permutation $P = (P_1, P_2, \dots, P_N)$ of $(1, 2, \dots, N)$ and length- M integer sequences $L = (L_1, L_2, \dots, L_M)$ and $R = (R_1, R_2, \dots, R_M)$.

For this permutation P , perform the following operation for $i = 1, 2, \dots, M$ in this order:

- Among $P_{L_i}, P_{L_i+1}, \dots, P_{R_i}$, swap the positions of the element with the minimum value and the element with the maximum value.

Find each element of P after the M operations.

Constraints

- $2 \leq N \leq 2 \times 10^5$
- $1 \leq M \leq 2 \times 10^5$
- P is a permutation of $(1, 2, \dots, N)$.
- $1 \leq L_i < R_i \leq N$
- All input values are integers.

Input

The input is given from Standard Input in the following format:

```
N M
P_1 P_2 ... P_N
L_1 R_1
L_2 R_2
⋮
L_M R_M
```

Output

Output each element of P after the M operations, in order from front to back, separated by spaces.

Sample Input 1

```
5 3
3 1 4 2 5
1 3
1 5
1 4
```

Sample Output 1

```
3 4 2 5 1
```

Initially, $P = (3, 1, 4, 2, 5)$.

- For $i = 1$: among P_1, P_2, P_3 , the minimum value is P_2 and the maximum value is P_3 . Swapping P_2 and P_3 gives $P = (3, 4, 1, 2, 5)$.
- For $i = 2$: among P_1, P_2, P_3, P_4, P_5 , the minimum value is P_3 and the maximum value is P_5 . Swapping P_3 and P_5 gives $P = (3, 4, 5, 2, 1)$.
- For $i = 3$: among P_1, P_2, P_3, P_4 , the minimum value is P_4 and the maximum value is P_3 . Swapping P_3 and P_4 gives $P = (3, 4, 2, 5, 1)$.

After the three operations, $P = (3, 4, 2, 5, 1)$.

Sample Input 2

```
6 7
3 6 5 2 4 1
4 5
2 3
3 5
4 6
3 4
1 6
3 5
```

Sample Output 2

```
3 5 4 6 2 1
```

F - Chebyshev Cafe

Time Limit: 3 sec / Memory Limit: 2048 MiB

Score : 525 points

Problem Statement

There is a city represented by a grid with N rows and N columns. The cell at the i -th row from the top and j -th column from the left is denoted as cell (i, j) . Each cell has one cafe.

You are planning a meeting. Among the participants, the number of people living in cell (i, j) is $(A_i \times B_j) \bmod M$. The travel cost for one participant living in cell (s_r, s_c) to go to the cafe at cell (t_r, t_c) is $\max(|s_r - t_r|, |s_c - t_c|)$.

Let $f(i, j)$ be the total travel cost incurred when gathering all the participants at the cafe at cell (i, j) .

For each cell (i, j) , find the following value, and output the bitwise XOR of these values over all cells.

- $f(i, j) + (i - 1)N + (j - 1)$

► What is the bitwise XOR operation?

Constraints

- $1 \leq N \leq 1500$
- $2 \leq M \leq 2 \times 10^6$
- $1 \leq A_i \leq M - 1 (1 \leq i \leq N)$
- $1 \leq B_j \leq M - 1 (1 \leq j \leq N)$
- All input values are integers.

Input

The input is given from Standard Input in the following format:

```
N M
A_1 ... A_N
B_1 ... B_N
```

Output

Output the answer.

Sample Input 1

```
4 19
2 3 1 1
3 4 4 10
```

Sample Output 1

```
467
```

The number of participants living in each cell is as follows:
$$\begin{bmatrix} 6 & 8 & 8 & 1 \\ 9 & 12 & 12 & 11 \\ 3 & 4 & 4 & 10 \\ 3 & 4 & 4 & 10 \end{bmatrix}.$$

For example, the value of $f(1, 2)$ is obtained as the sum 176 of the following 16 values:

- The total amount $6 \times 1 = 6$ paid by the 6 residents of cell $(1, 1)$, each of whom pays a travel cost of $\max(|1 - 1|, |1 - 2|) = 1$
- The total amount $8 \times 0 = 0$ paid by the 8 residents of cell $(1, 2)$, each of whom pays a travel cost of $\max(|1 - 1|, |2 - 2|) = 0$
- The total amount $8 \times 1 = 8$ paid by the 8 residents of cell $(1, 3)$, each of whom pays a travel cost of $\max(|1 - 1|, |3 - 2|) = 1$
- The total amount $1 \times 2 = 2$ paid by the 1 resident of cell $(1, 4)$, each of whom pays a travel cost of $\max(|1 - 1|, |4 - 2|) = 2$
- $\vdots$
- The total amount $3 \times 3 = 9$ paid by the 3 residents of cell $(4, 1)$, each of whom pays a travel cost of $\max(|4 - 1|, |1 - 2|) = 3$
- The total amount $4 \times 3 = 12$ paid by the 4 residents of cell $(4, 2)$, each of whom pays a travel cost of $\max(|4 - 1|, |2 - 2|) = 3$
- The total amount $4 \times 3 = 12$ paid by the 4 residents of cell $(4, 3)$, each of whom pays a travel cost of $\max(|4 - 1|, |3 - 2|) = 3$
- The total amount $10 \times 3 = 30$ paid by the 10 residents of cell $(4, 4)$, each of whom pays a travel cost of $\max(|4 - 1|, |4 - 2|) = 3$

The value of $f(i, j)$ for each cell is as follows:
$$\begin{bmatrix} 220 & 176 & 179 & 224 \\ 199 & 140 & 136 & 182 \\ 212 & 159 & 143 & 178 \\ 255 & 215 & 201 & 218 \end{bmatrix}.$$

Sample Input 2

```
6 67
1 6 9 2 3 1
1 10 3 7 2 2
```

Sample Output 2

```
1021
```

Sample Input 3

```
20 676767
60932 508200 22883 481150 426836 554098 276223 307959 133025 573906 17711 413230 274246 500
277 524959 477728 546547 43099 324227 660187
552697 293978 246000 159233 213736 628198 394200 194076 618720 453035 615100 331522 669419
180856 530693 399531 223149 159334 102107 598633
```

Sample Output 3

```
764668547
```

G - Increasing Popcount

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 625 points

Problem Statement

You are given positive integers L and R satisfying $L \leq R$.

A sequence of positive integers $A = (A_L, A_{L+1}, \dots, A_R)$ of length $R - L + 1$ satisfying the following condition is called a **good integer sequence**:

- For every pair of integers (i, j) satisfying $L \leq i < j \leq R$, if $A_i = A_j$, then $\text{popcount}(i) < \text{popcount}(j)$.

Find the minimum value of $\max(A_L, A_{L+1}, \dots, A_R)$ over all good integer sequences $A = (A_L, A_{L+1}, \dots, A_R)$.

You are given T test cases; solve each of them.

► What is popcount?

Constraints

- $1 \leq T \leq 10^4$
- $1 \leq L \leq R \leq 10^{18}$
- All input values are integers.

Input

The input is given from Standard Input in the following format:

```
T
case1
case2
⋮
caseT
```

Each test case is given in the following format.

```
L R
```


Output

Output T lines.

The i -th line ($1 \leq i \leq T$) should contain the answer for the i -th test case case_i .

Sample Input 1

```
5
3 6
1 1
2 15
100 100000000
514284786278117031 620546740167642909
```

Sample Output 1

```
3
1
6
1636985
11375123021856567
```

Consider the first test case.

$(A_3, A_4, A_5, A_6) = (1, 2, 2, 3)$ is a good integer sequence, and $\max(A_3, A_4, A_5, A_6) = 3$.

There is no good integer sequence with $\max(A_3, A_4, A_5, A_6) < 3$, so output 3 on the first line.

Hints: AtCoder Beginner Contest 476

Taken from the official editorial, not written by us. Read one hint at a time.

A. Appender *(editorial, first line)*

Check if the last character of the given string coincides with e, append an appropriate string to S , and print it.

B. Wild Card *(editorial, first line)*

If S matches T , then for $i = 1, 2, \dots, N$, T_i equals either S_i or $*$.

C. Third Largest Number *(editorial, first line)*

Consider finding the answer in ascending order of k .

D. Automat *(editorial, first line)*

A combination of products $A_{i_1}, \dots, A_{i_n}, B_{j_1}, \dots, B_{j_m}$ can be purchased if and only if: the total price, $\sum_{1 \leq k \leq n} A_{i_k} + \sum_{1 \leq k \leq m} B_{j_k} \dots$

E. Min-Max Swap *(editorial, first line)*

Consider simulating the operations step-by-step.

F. Chebyshev Cafe *(editorial, first line)*

We will find the values of $f(i, j)$ for all N^2 cells.

G. Increasing Popcount *(editorial, first line)*

First, consider a graph with $(R - L + 1)$ vertices, vertex L through R .

Source: <https://atcoder.jp/contests/abc476/editorial>